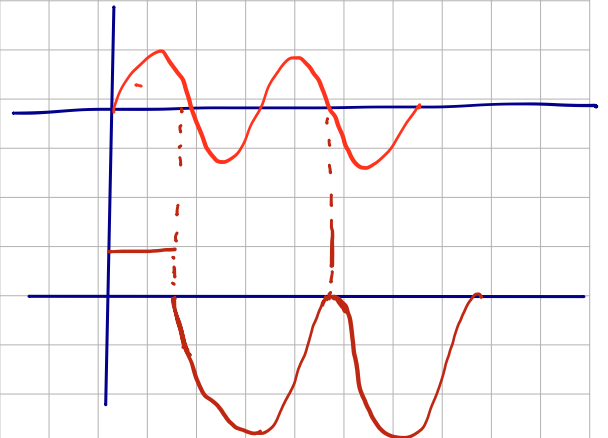
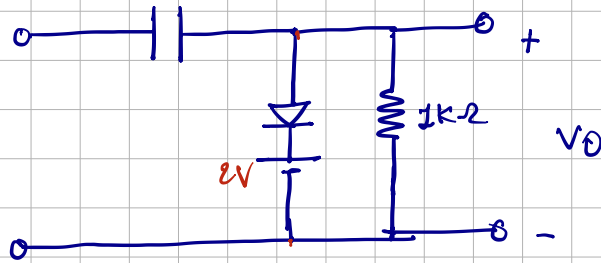


[WEEK-7 Graded assignment]

Roll No :- 25F3100145

NAME :- SHARU ROHIT KAMLESH

Q.1]
Soln



This represent a negative clamper circuit.

For positive half cycle diode will conduct & hence V_o will be zero.

Since there is battery connected hence capacitor will discharge upto 2 volts only

∴ Net peak voltage is $-10 - 8 = -18V$

∴ therefore option 4 is correct

Q.2]

⇒ maximum power = 300 mW

$$\therefore V \cdot I = 300 \text{ mW}$$

$$I = \frac{300 \text{ mW}}{6 \text{ V}}$$

$$\boxed{I_{\text{max}} = 50 \text{ mA}} \text{ through diode.}$$

Q.3)
Soln:- C_1 is true since clamper circuit shifts the waveform
 C_2 is also true since it doesn't change the waveform

Q.4)
Soln:- Given circuit is a positive clamper circuit.
Since there is a battery added to the circuit
Capacitor will only charge upto $= 20V$
& hence we get maximum output $= 20 + 30$
 $= 50V$

Q.5)
Soln:- Since Zener breakdown voltage is $7V$
& $7V > \text{input}$
Zener diode will act as open circuit,
hence no current will flow & hence
'0' output voltage.

Q.6]

Soln:-

Given Circuit is negative clamper circuit.

During positive half cycle diode will conduct
& hence capacitor will charge upto $-13V$

& then diode will be reverse biased

& output will further increase during negative
half cycle.

$$\therefore -13 - 10 = -23V \text{ maximum.}$$

Q.7]

Soln:- ① During positive half cycle diode will not conduct &
we get -3 volts across V_o

② During negative half cycle diode will conduct &
capacitor will charge to value of $10 - 3 = 7V$

③ when input again start to increase, capacitor will
start to discharge & voltage further rises
to $7 + 10 = 17$

Q.8]

Soln:- ① In this circuit for $T/2$ period we get $3V$ as V_o

Since diode will be reverse bias

② After $T/2$ diode will start to conduct & we capacitor
will charge to value $3 + 10 = 13$ volts.

③ And then again for positive half cycle diode will be
reverse bias & voltage will be keep adding till $13 + 10 = 23V$

Q.9]

Soln:- $V_{BE} = 0.7 \text{ V}$

$$V_B = \frac{24 \times 9}{36+9} = 4.8 \text{ V}$$

Emitter voltage $V_E = V_B - V_{BE} = 4.1 \text{ V}$

$$\therefore I_E = \frac{V_E}{R_E} = \frac{4.1}{2\text{K}} = 2.05 \text{ mA}$$

$$\begin{aligned} \text{Collector current} &= \frac{\beta}{\beta+1} I_E = \frac{100}{101} \times 2.05 \\ &= 2.029 \text{ mA} \end{aligned}$$

Q.10]

$\Rightarrow$

$$V_C = V_{CC} - I_C R_C$$

$$= 24 - (2.029 \times 9)$$

$$= 5.73 \text{ V}$$

& $V_E = 4.1 \text{ V}$

$$\therefore V_{CE} = 5.73 - 4.1$$

$$= 1.63$$